

# Compare GPT-3.5 Turbo and GPT-4 Turbo at Lowest word-level confidence experiment

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# Idea

Based on last experiment that we consider sentence-level confidence, Mathew suggested me to do word-level confidence experiment.

Enno suggested me that it could be easier to start with lowest word-level confidence.

For the moment, I did two experiments with 2 model:

gpt-3.5-turbo-1106 and gpt-4-1106-preview without considering lowest word-level confidence.

# Results for GPT-4 Turbo vs GPT-3.5 Turbo

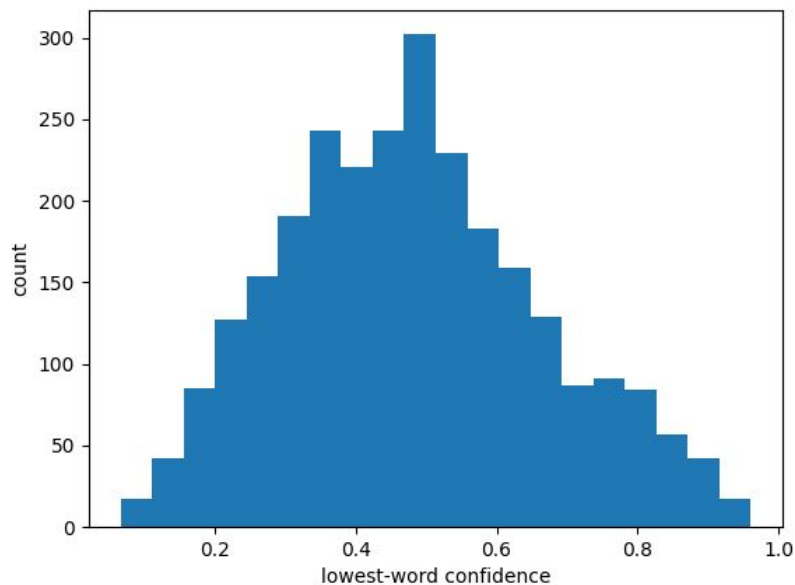
## GPT-4 Turbo

```
WER_original:      8.6116%  
WER_corrected_chatgpt: 6.9935%  
CER_original:      3.5707%  
CER_corrected_chatgpt: 3.4570%  
SER_original:      61.4502%  
SER_corrected_chatgpt: 56.0488%  
---
```

## GPT-3.5 Turbo

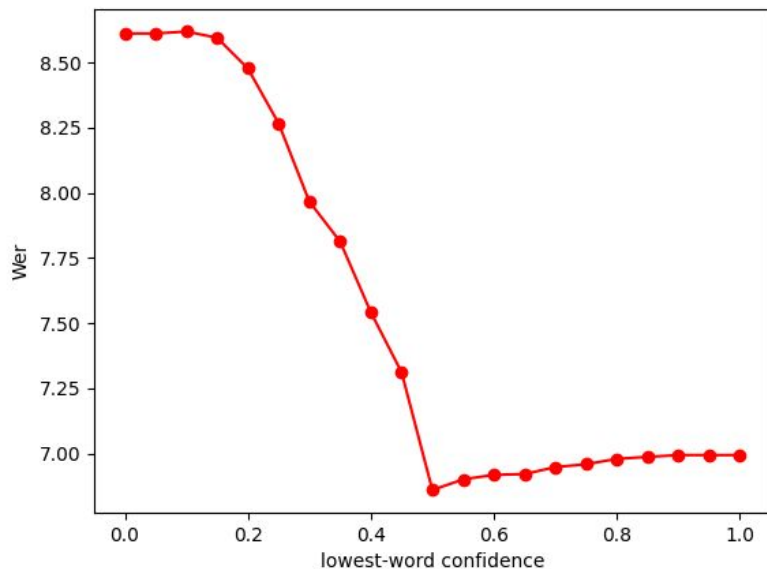
```
WER_original:      8.6116%  
WER_corrected_chatgpt: 6.8042%  
CER_original:      3.5707%  
CER_corrected_chatgpt: 3.1879%  
SER_original:      61.4502%  
SER_corrected_chatgpt: 54.7170%  
---
```

# Histograms for GPT-4 Turbo and GPT-3.5 Turbo

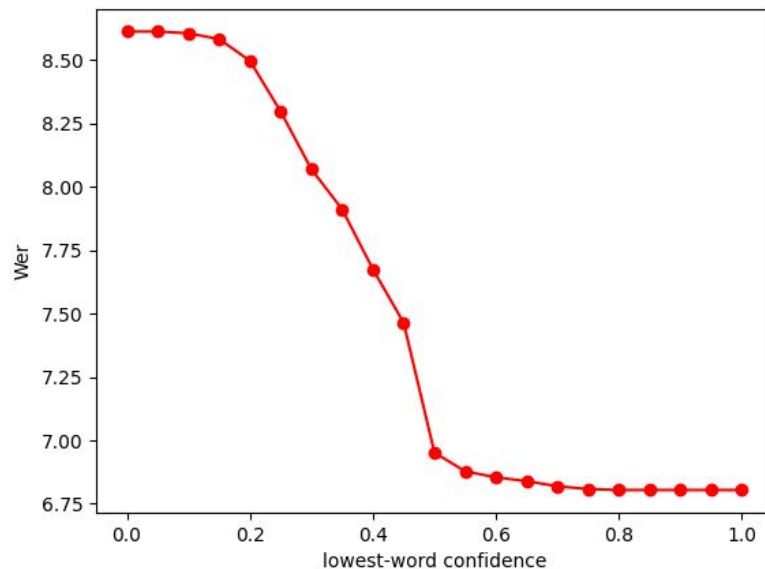


# Wer plots for GPT-4 Turbo and GPT-3.5 Turbo

GPT-4 Turbo



GPT-3.5 Turbo



# Results

Based on Wer vs lowest-word confidence plot, we can say that gpt-3.5-turbo-1106 is more powerful than gpt-4-1106-preview in reducing the wer error at ASR outputs for tiny whisper model.

And for gpt-3.5-turbo-1106, we can see that we have a threshold near 0.6 at Wer vs lowest-word confidence plot.

# My team's suggestions about this experiment

Mathew and Enno suggested me to try different whisper models (small-medium-large) for both gpt-3.5-turbo-1106 and gpt-4-1106-preview and compare all the results of these model together.

Mathew and Enno mentioned that the lowest-word level confidence suffices for the moment, and there's no need for another experiment regarding word-level confidence.

Mathew also suggested me for next step we can use LF-MMI or Hybrid-HMM/ANN instead of whisper model.



Thank you